

Cloud Computing Architecture

Module 37

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Hello friends I welcome you all to the video lecture series of cyber security and information security in this lecture we will study cloud computing architecture first of all please see the contents which I shall cover after covering brief introduction about cloud computing I shall take up essential characteristics of cloud which will be followed by three Services offering model and last I shall take up Cloud architecture in the simple terms cloud computing means storing and accessing data and programs over the internet from a remote location or computer instead of our computer's hard drive it is called remote location and has several properties such as scalability elasticity Etc which is significantly different from a simple remote machine the cloud is just a metaphor for

the internet when we store data on or run a program from the local computer's hard drive that is called local storage and Computing for it is to be considered cloud computing we need to access our data or programs over the Internet the end result is the same however with an online connection cloud computing can be done anywhere anytime and by any device in cloud computing the it and business resources such as servers storage Network applications and processes can be dynamically provisioned to the user needs and workload in addition while a cloud can provision and support a grid a cloud can also support non-grid environments such as three TIRE web architecture running on traditional or web to Applications therefore cloud computing

is a mechanism of bringing hiring or getting the services of the computing power or infrastructure to an organizational or individual level to the extent required and paying only for the consumed services in addition one can think to add security while accessing these remote Computing resources as depicted in the figure figure shows several cloud computing applications the cloud represents the internet based Computing resources and the accessibility is through some secure support of connectivity 3 minutes, 1 second it is a Computing solution growing in popularity especially among individuals and small and mediumsized companies thus cloud computing comes into focus and much needed only when we think about what Computing resources and

Information Technology Solutions are required this need caters to a way to increase capacity or add capabilities on the fly without investing in new infrastructure training new personal or lensing new software essential characteristics cloud computing has five essential characteristics listeners can note the word essential which means that if any of these characteristics is missing then it is not cloud computing on demand self-service it has the flexibility of self-provisioning its resources according to the demands of the customer without the need of being aided by any human being in network access it makes possible for wide range of heterogeneous Network accessibility for mobile clients as well as other

common computer models broad network access you can access these resources from anywhere and you can access them from a browser from a desktop with applications designed to work with them or from from a mobile device one of the most popular applications model such as iPhone apps is a mobile application that communicates with the cloudbased patent elastic resource pooling the providers Computing resources are pooled to serve multiple consumers using a multi-tenant model with different physical and virtual resources dynamically assigned and reassigned according to Consumer demand there is a sense of location Independence in that the customer generally has no control or knowledge over the exact location of the provided

resources but may be able to specify the location at a higher level of abstraction examples of resources include storage processing memory and network bandwidth rapid elasticity it has the capabilities of managing the pools of resources on demand basis when need to and free up the resources as situation demands location independent resource pooling the resources for the customers are aggregated in multi-tenant model irrespective of the geolocation of the consumers the customers have no idea about the location of the resources he she were provision from except some abstract high level of information like countries and regions to the consumer the capabilities available for provisioning often appear to be

unlimited and can be purchased in any quantity at any time Meed service cloud computing has a metering capability which controls monitors and reports the use of resources this results in audited reports for both the provider and the consumer showing exec how much hard disk was used which Services were accessed and for how many hours the implication for companies that use cloud computing means they pay only for what they use which is very coste efficient for business operations resource uses can be monitored controlled and reported providing transparency for both the provider and consumer of the utilized service cloud depl deployment 7 minutes, 1 second models deployment models describe the ways with which the cloud services can be deployed or made available to its

customers depending on the organizational structure and the provisioning location Cloud that is internet based Computing resources that is the locations where data and services are acquired and provision to its customers can take various forms for deployment models are public Cloud private Cloud Community cloud and hybrid Cloud private Cloud it is also known

as internal Cloud the platform for cloud computing is implemented on the cloud-based secure environment that is safeguarded by a firewall which is under the governance of the IT department that belongs to the particular corporate private Cloud as it permits only the authorized users has direct control over their data what exactly constitutes a private Cloud it is difficult to Define

because when it's classified according to the services there are significant variations whether the physical computers are hosted internally or externally they provide the resources from a distinct pool to a private cloud services business reses that have Dynamic needs assignments which are mission critical Security Alarms management demands and uptime requirements are better suited to adopt private Cloud obstacles with regards to security can be evaded in a private Cloud that in case of natural disaster and internal Data Theft the private Cloud may be prone to vulnerabilities public Cloud public cloud is a type of of cloud hosting in which the cloud services are delivered over a network which is open for public usage this model is a true

representation of cloud hosting the customers do not have any distinguished ability and control over the location of the infrastructure from the technical Viewpoint there may be slight or no difference between private and public Cloud's structural design except in the level of security offered for various Services given to the public Cloud subscribers by the cloud hosting providers public cloud is better suited for business requirements which require managing the load host application that is saas based and manage applications that many users consume due to the decreased capital overheads and operational cost this model is economical the dealer May provide the service free or in the form of the license policy like pay per user the

cost is shared by all the users so public Cloud profits the customers more by achieving economies of scale public Cloud facilities may be availed free example a public cloud is Google Community Cloud it is a type of cloud hosting in which the setup is mutually shared between many organizations that belong to a particular community that is Banks and trading firms it is a multi setup that is shared among several organizations that belong to a specific group which has a similar Computing apprehensions the community members generally share similar privacy performance and security concerns the main intention of these communities is to achieve their business related objectives a Community Cloud may be

internally managed or it can be managed by a third party provider it can be hosted externally or internally the cost is shared by the specific organizations within the community hence Community cloud has cost saving capacity organizations have understood that cloud hosting has a lot of potential so selection of the right type of cloud hosting is needed thus you need to know your business and analyze the demands once the appropriate type of cloud hosting is selected you can achieve your business related goals more easily you can channelize all your efforts to take those strategic steps that will help your business to succeed hybrid Cloud it is a type of cloud computing which is integrated it can be an arrangement of two or more Cloud servers that is

private public or Community Cloud that is bound together but remain individual entities. Benefits of the multiple deployment models are available in hybrid Cloud hosting. A hybrid Cloud can cross isolation and overcome boundaries by the provider hence it cannot be simply categorized into public private or Community Cloud. It permits the user to increase the capacity or the capability by aggregation, assimilation or customization with another Cloud package or service. In a hybrid Cloud, the resources are managed and provided either inhouse or by external providers. It is an adaptation among two Platforms in which the workload exchanges between the private cloud and the public Cloud as per the need and demand. Resources that are non critical like development and test workloads can

be housed in the public Cloud that belongs to a third party provider while the workload set are critical or sensitive must be housed internally. Consider an e-commerce website which is hosted on a private Cloud that gives security and scalability since security is not a prime concern for those broker site it is hosted on a public Cloud which is more economical as compared to a private Cloud. Businesses that have more focus on security and demand for their unique presence can implement hybrid Cloud as an effective business strategy when facing demand spikes. The additional resources that are required by a particular application can be accessed from the public Cloud. This is termed as Cloud bursting and is available with the hybrid

Cloud organizations can use the hybrid cloud model for processing big data on a private cloud. It can retain sales business and various data and can initiate analytical queries over the public Cloud as the public cloud is effective to meet the demand spikes. Hybrid cloud hostings is enabled with features like scalability, flexibility and security if one is ready to overlook a few challenges like application program interface, compatibility, network connectivity issues and capital expenditures. Then hybrid Cloud would be an appropriate option. Three service offering models: the three kinds of services with which the cloud-based Computing resources are available to end customers are software as a service

SaaS, platform as a service PaaS and infrastructure as a service IaaS. It is also known as the service platform in infrastructure SaaS model. In 15 minutes, 1 second the cloud and it is shown in the figure. SaaS is a software distribution model in which applications are hosted by a vendor or service provider and made available to customers over a network. PaaS is a paradigm for delivering operating systems and Associated Services. Example: computer AED software engineering CASE tools, integrated development environments, IDEs for developing software. Solutions over the internet without downloads or installation. IaaS involves Outsourcing the equipment used to support operations including storage, Hardware servers and networking components. Clouds are

the capability provided to the consumer is to use the provider's applications running on a cloud infrastructure including network servers, operating systems, storage and even individual application capabilities with the possible exception of limited user specific application

configuration settings the applications are accessible from various client devices through either A Thin Client interface such as web browser or a program interface the consumer does not manage or control the underlying Cloud infrastructure typical applications offered as a service include customer relationship management CRM business intelligence analytics and online accounting software Cloud paaS the capability provided to Consumer

is to deploy onto the Cloud infrastructure the consumer created or acquired applications created using programming languages libraries services and tools supported by the provider the consumer does not manage or control the underlying Cloud infrastructure but has control over the deployed applications and possibly configuration settings for the application hosting environment in other words it is a packaged and ready to run development or operating framework the PaaS window provides the networks servers storage and manages the levels of scalability and maintenance the client typically pays for services used examples of PaaS providers include Google App Engine and Microsoft Azure Services Cloud IaaS the capability provided to the consumer

is to provision processing storage networks and other fundamental Computing resources on a pay per use basis where he or she is able to deploy and run software which can include operating systems and applications the consumer does not manage or control the underlying Cloud infrastructure but has control over the operating systems storage and deployed applications and possibly limited control of select networking components the service provider owns the equipment and is responsible for housing cooling operation and maintenance Amazon web services AWS is a popular example of a large IAS provider the major difference between PaaS and IAS is the amount of control that you users have in essence PaaS allows

vendors to manage everything while IAS requires more management from the customer site generally speaking organizations that already have a software package or application for a specific purpose and want to install and run it in on the cloud should opt to use IAS instead of PaaS so now we come over to the cloud architecture the cloud has high level view of an architecture that describes its working mechanism it includes the dependencies on which it works and the components that work over it the cloud is a recent technology that is completely dependent on the internet for its functioning figure depicts the architecture the cloud architecture can be divided into four layers based on the access of the cloud by the user they are

as follows layer one that is user client layer this layer is the lowest layer in the cloud architecture all the users or client belong to this layer this is the place where the client user initiates the connections to the cloud the client can be any device such as a Thin Client thick client or mobile or any handheld device that would support basic functionalities to access a web application the Thin client here refers to a device that is completely dependent on some other system for its complete functionality in simple terms they have very low processing capability similarly thick clients are General computers that have adequate processing capability they have sufficient capability for independent

work usually a cloud application can be accessed in the same way as web application but internally the properties of cloud applications are significantly different thus this layer consists of a client devices Layer Two that is Network layer this layer allows the users to connect to the cloud the whole Cloud infrastructure is dependent on this connection where the services are offered to customers this is primarily the internet in the case of a public Cloud the public Cloud usually exists in the specific location and the user would not know the location as it is abstract and the public Cloud can be existed all over the world in the case of a private Cloud the connectivity may be provided

by a local area network even in this case the cloud completely depends on the network that is used usually when accessing the public or private Cloud the users require minimum bandwidth which is sometimes defined by the cloud providers this layer does not come under the purview of a service level agreements SLAs SLAs do not take into account the internet connection between the user and Cloud for quality of service that is QoS layer three that is cloud management layer this layer consists of softwares that are used in managing the cloud these softwares can be Cloud operating system OS which acts as an interface between the data center and the user or a management software that allows managing resources these softwares usually allow

Resource Management that is scheduling provisioning Etc optimization server consolidation storage workload consolidation and internal Cloud governance this layer comes under the purview of SLAs that is the operations 24/7 minutes, 1 second taking place in this layer would affect the SLAs that are being decided upon between the users and service providers any delay in processing or any discrepancy in service provisioning may lead to an SLA violation as per rules any SLA violation would result in penalty to be given by the service provider these SLAs are for both private and public clouds popular service providers are Amazon web services AWS and Microsoft Azure for public Cloud similarly open stack allow private Cloud

creation deployment and management layer four Hardware resource layer layer four consists of Provisions for actual Hardware sources usually in the case of a public Cloud a data center is used in the back end similarly in a private Cloud it can be data center which is a huge collection of Hardware sources interconnected to each other that is present in a specific location or a high configuration system this layer comes under the purview of SLAs this is the most important layer that governs the SLAs this layer affects the SLAs most in the case of data centers whenever a user accesses the cloud it should be available to the users as quickly as possible and should be within the time that is defined by the SLAs if there is any discrepancy in

provisioning the resources or applications the service provider has to pay the penalty hence the data center consist of high speed network connection and a highly efficient algorithm to transfer the data from the data center to the manager there can be a number of data centers for a cloud and similarly a number of clouds can share a data center thus this is the architecture of a cloud there can be little loose isolation between layer three and

layer four depending on the way the cloud is deployed so friends this was all about cloud computing and its architecture hope the concepts explained in this lecture were understandable and helpful hope to see you in the next lecture till then goodbye goodbye thank you